Project Blocker & Technical Risk Report

Report Type: Impediments & Conceptual Hurdles **Date:** 05-Sep **Target Track:** EDA & Algorithmic Mathematics **ACTION REQUIRED**

BLOCKER 1: EXPLORATORY DATA ANALYSIS (EDA) UNDERSTANDING & PIPELINE HANDOFF

⚠ Challenge: Translating Raw EDA Insights into Feature Engineering & ML Inputs

● PRIORITY: HIGH

- **Feature Distribution & Imbalance:** Difficulty visualizing and interpreting class skew in `Loan_Approval_Status`, directly explaining why KNN displayed low recall (31.25%) while Decision Tree captured 60.42% of approvals.
- **Multicollinearity & Correlations:** Interpreting correlation matrices between financial attributes (`Annual_Income`, `Monthly_Expenses`, and `Loan_Amount_Requested`) to avoid redundant splits in tree modeling.
- **Categorical Encoding Impact:** Grasping how expanding `Gender`, `Loan_Purpose`, and `Loan_Type` into 13 one-hot columns affects the dimensionality and sparsity of the dataset.

Resolution & Next Steps: Keerti C and Aditya K will schedule a walkthrough with the modeling team to review automated pairplots, correlation heatmaps, and target distribution charts before model deployment.

BLOCKER 2: MATHEMATICAL FORMULATIONS IN DECISION TREES & KNN

⚠ Challenge: Conceptualizing Loss Functions, Impurity Metrics, and Distance Calculations

● PRIORITY: MEDIUM-HIGH

Model Track	Mathematical Concept	Project Implication
Decision Tree (CART)	**Gini Impurity:** `Gini = 1 - ∑ (p_i)^2`	Explains why **Credit_Score (30.42%)** was chosen as the root split: it achieved the maximum Gini gain by isolating high-probability non-default classes.

K-Nearest Neighbors	**Euclidean Distance & Scaling:**
	`d(p, q) = √∑(p_i - q_i)^2`
	`z = (x - μ) / σ`
		Without `StandardScaler`, `Annual_Income` (scale: 10,000s) would dominate Euclidean distance over `Loan_Term` (scale: 10s), distorting nearest neighbors.

Classification Metrics	**Precision vs. Recall Trade-Off:**
	`Precision = TP / (TP + FP)`
	`Recall = TP / (TP + FN)`
		Explains the trade-off: KNN achieved high precision (78.95%) with only 4 FP, whereas Decision Tree achieved high recall (60.42%) by capturing 29 approvals.

Resolution & Next Steps: Abdussamad S and Namrata D will integrate interactive visual aids (tree schematics and scaled neighbor plots) into the final presentation deck to clarify these concepts for technical and non-technical stakeholders.

SUMMARY & ACTION PLAN

- **Deliverable 1:** Host a 30-minute cross-track review between EDA coordinators and model developers to finalize feature interpretations.
- **Deliverable 2:** Include an annotated "Mathematics Behind the Models" reference slide in the project slide deck before final submission.